

Supplemental Material for

Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning

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Abstract

As supplemental material for our review article “Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning”, we provide a summary—in table form—of the 84 approaches mentioned in Section 3, with a specific focus on *(i)* the learning models used, *(ii)* the safety properties achieved, and *(iii)* the robotic tasks considered.

SUMMARY OF REVIEWED LITERATURE

Supplemental Table 1 summarizes the safe learning control approaches reviewed in our paper “Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning”. Learning approaches are classified as *Model-Based* if a dynamics model is used to produce control inputs, or *Model-Free* if there is a direct mapping from states or measurements to control inputs. For model-based approaches, we categorize the structure of the *A Priori* and the *Learned* component of the model as Gaussian process (*GP*), neural network (*NN*), or *Other* for non-standard methods (e.g., set computation from data).

The *Safety Properties* category highlights which algorithms can handle state and/or input constraints (*Constraint Satisfaction*) and which ones provide stability guarantees (*Stability*). For both of these properties, we report whether they are enforced while the policy is being learned (*During Learning*) or only by the final, learned policy (*After Learning*). The *Robustness* category indicates whether the final policy is designed to be robust to *Input* disturbances, model *Parameter* uncertainty, and/or other *Dynamics* uncertainties broader than the previous two types of disturbances.

Finally, the last category reports the *Task(s)* each method was applied to: *Stabilization* for stabilizing an equilibrium of the system; *Tracking* for tracking of a given trajectory; *Navigation* for navigating or planning a sequence of actions to reach a given goal; *Locomotion* for the movement of legged robots like humanoids or quadrupeds; and *Manipulation* for pushing, reaching, or grasping operations using a robotic arm.

Supplemental Table 1: Summary of the key properties of the works referenced in Section 3 of the review article.

Section		Learning			Safety Properties					Task
		Model-based		Model-free	During Learning		After Learning			
		<i>A Priori</i>	Learned		Constraint Satisfaction	Stability	Constraint Satisfaction	Stability	Robustness w.r.t.	
Learning Uncertain Dynamics Sec. 3.1	Sec. 3.1.1	Linear (37, 38, 42, 43), Nonlinear (40, 41)	NN (37, 42, 43), GP (38, 40, 41)			(37, 38, 40, 41, 42, 43)		(37, 38, 40, 41, 42, 43)	Input (37, 38), Dynamics (40, 41, 42, 43)	Tracking (37, 38, 40, 41, 42, 43)
	Sec. 3.1.2	Linear (44, 45, 46), Nonlinear (47, 48)	GP (44, 46, 47, 48) Other (45)			(44, 46, 47, 48)		(44, 45, 46, 47, 48)	Dynamics (44, 45, 46, 47, 48)	Stabilization (44, 45, 46), Tracking (47, 48)
	Sec. 3.1.3	Linear (49, 50, 51, 52, 56, 57, 58, 59), Nonlinear (53, 54, 60, 61, 62, 63, 64, 65)	NN (58, 64), GP (59, 60, 61, 62, 63), Other (49, 50, 51, 52, 53, 54, 56, 57, 65)		Input & state (49, 50, 51, 52, 53, 54, 56, 57, 58, 59, 60, 61, 62, 63, 64)	(49, 50, 51, 52, 53, 54, 56, 57, 58, 59, 60, 61, 62, 63, 64)	Input & state (49, 50, 51, 52, 53, 54, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65)	(49, 50, 51, 52, 53, 54, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65)	Input (57), Dynamics (58, 59, 60, 61, 62, 63, 64, 65), Parameters (49, 50, 51, 52, 53, 54, 56, 57)	Stabilization (50, 51, 52, 53, 54, 56, 57, 58, 59, 62, 63), Tracking (49, 60, 61, 64, 65)
	Sec. 3.1.4	Nonlinear (66)	GP (66)			(66)		(66)		Stabilization (66)
Encouraging Safety in RL Sec. 3.2	Sec. 3.2.1	Nonlinear (67, 77, 80),	GP (76, 77, 78, 79)	NN (68, 73, 81, 82, 83), Tabular (71)	Input & state (67, 68, 71, 73, 76, 77, 78, 79, 80) Trajectory (83)		Input & state (67, 68, 71, 73, 76, 77, 78, 79, 80) Trajectory (81, 82, 83)			Stabilization (79), Tracking (77), Navigation (67, 68, 71, 80, 81, 82), Locomotion (81, 83), Manipulation (73, 81, 82, 83)
	Sec. 3.2.2	Nonlinear (85, 86)	NN (85, 86, 87, 89)	NN (90)			Input & state (85, 86, 87, 89)		Dynamics (90)	Stabilization (87), Navigation (85, 86, 87, 89), Locomotion (87, 90), Manipulation (89)
	Sec. 3.2.3	Tabular (94, 96)		NN (32, 92, 94, 95, 96)	Trajectory (32, 94, 95, 96)		Trajectory (32, 92, 94, 95, 96)			Navigation (32, 92, 94, 95, 96), Locomotion (32, 95, 96)
	Sec. 3.2.4		NN (103)	NN (34, 35, 36, 101, 102, 104, 105)	Input & state (103)	(103)	Input & state (103)	(103)	Input (35, 36), Dynamics (34, 101, 102, 103), Parameters (104, 105)	Stabilization (34, 101, 105), Tracking (103), Navigation (101, 102), Locomotion (34, 35, 36, 104), Manipulation (105)
Certifying Learning- based Control Sec. 3.3	Sec. 3.3.1	Linear (107), Nonlinear (106, 108, 110, 111)	NN (106, 107, 108, 110), Other (111)			(107)		(106, 107, 108, 110, 111)	Dynamics (106, 107, 108, 110, 111)	Stabilization (110, 111), Tracking (106, 107, 108, 111), Locomotion (111)
	Sec. 3.3.2	Linear (136, 137), Nonlinear (116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 130, 131, 132, 138, 139)	GP (118, 125, 127, 128, 130, 137), NN (116, 117, 119, 120, 121, 125, 126, 132), Other (122, 123, 124, 131, 136, 137, 138)	NN (118, 119, 133)	Input & state (120, 121, 122, 123, 124, 125, 126, 127, 136, 137, 138, 139), State (130, 131, 132)	(120, 121, 122, 123, 124, 125, 126, 127, 130, 131, 132, 136, 137, 138)	Input & state (116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 136, 137, 138, 139), State (130, 131, 132, 133)	(116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 130, 131, 132, 133, 136, 137, 138)	Dynamics (116, 117, 118, 119, 120, 121, 122, 125, 126, 127, 130, 131, 132, 133, 136, 137, 138), Parameters (123, 124, 138)	Stabilization (125, 133, 136, 138, 139), Tracking (116, 117, 120, 121, 122, 123, 124, 126, 127, 128, 130, 131, 137), Navigation (118, 132), Locomotion (119)

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